Mapping Climate Extremes and Economic Vulnerabilities across the United States

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1 Introduction

The number and severity of catastrophic weather events (e.g., hurricanes, wildfires, floods) have been steadily increasing as we experience the warmest global period on record (Arias et al. (2021); CarbonBrief (n.d.); Parks et al. (2023)). The Climate Risk Index 2025 lists the United States as the seventh most affected country, experiencing more than 500 fatalities and approximately 140 billion USD losses resulting from extreme weather events in 2022 alone (Adil et al. (2025)). Climate-change-related weather events have been self-reported to have significantly affected 48% of US residents, with 33% reporting that they will likely be displaced from their homes because of climate events within the next 25 years (World Economic Forum et al. (2022)). Moreover, home ownership rates are likely to fall in areas where a natural disaster occurred in recent years (Sheldon and Zhan (2019)), a trend influenced by a variety of factors, including post-disaster changes in housing prices and income (Murphy and Strobl (2009)), and variability in federal aid disbursal in the recovery period after a significant weather event (Giordono et al. (2025); Waters et al. (2024)). Given these disparities, it is important to consider weather-related risk factors and their impact on socioeconomic factors over time when determining where to live and how to govern. Indeed, U.S. News & World Report's Best Places to Live rankings include environmental risk as a key factor alongside other metrics such as median income, cost of living, affordability of housing, and unemployment (U.S. News & World Report (2025)). But metrics only go so far when it comes to communicating this kind of message to a vast audience. Our project uniquely integrates multi-hazard data and economic recovery metrics into a single index, and displays it intuitively on an interactive map of the U.S. that enables engagement and sparks discussion.

2 Problem Definition

It is difficult for the U.S. public and government to understand the complete picture of the damage caused by extreme weather across different regions. At present, policymakers, researchers, or members of the public cannot easily track the compounded impact of multiple weather hazard types over time at a consistent, high-resolution (county-level) geography. Therefore it is difficult to correlate these environmental events with human impacts like job security, and to compare communities' economic danger against the federal aid it receives.

Weather event records, federal disaster aid obligations, and economic metrics are maintained in separate databases. For example, National Oceanic and Atmospheric Administration's (NOAA's) Storm Events Database provides comprehensive detail while Federal Emergency Management Agency's (FEMA's) Funding Aid data is collected at disaster declaration level, and insurance and housing equity statistics are updated at frequent time lags. Our project formally defines the problem as follows: Given county-level time-series data on weather events, disaster aid, housing markets, and unemployment, create an interactive choropleth visualization summarized by an easy-to-consume risk index. Such an index would help users understand vulnerability of geographic locations, and make adjustments to the allocation of federal aid.

3 Literature Survey

The concept of measuring climate hazards risk and risk is well established. Narayanan et al. (2024) introduced the Climate Extremes Resilience Index (CERI) which combines the revised version of the U.S. Climate Extremes Index (CEI) with Baseline Resilience Indicators for Communities (BRIC) to assess county-level potential impacts of climate extremes. However, CERI only captures short-term impacts and does not account for cascading secondary effects, for example prolonged health burdens, long-term migration, infrastructure. Several studies examine how extreme weather affects public health. Zhang et al. (2024) showed how rainfall and snow reduce emergency medical access in Beijing, while Salas et al. (2024) linked post-disaster hospitalization and mortality trends to counties suffering the highest economic losses. Robinson (2021), Giordono et al. (2025), and Waters et al. (2024) reviewed U.S. extreme weather, policy response, and aid disparities—highlighting gaps between reported damages and obligated federal assistance. Forecasting and modeling studies such as Kan et al. (2025) and Shenoy et al. (2022) demonstrate methods for predicting heatwaves and modeling long-term shifts in weather regularity. These studies focus on niche hazards and fail to recommend a related policy goal, unlike our project's multi-hazard, easy-to-consume approach. Howe et al. (2014) and Allan et al. (2020) explored how personal experience and perception shape climate awareness; our project builds on these findings to visualize actual, data-driven risks. Research also connects extreme weather to economic outcomes. Murphy and Strobl (2009) and Kousky et al. (2020) examined housing markets under hurricane and flood risks, while Sheldon and Zhan (2019) studied home-ownership shifts in disaster-prone regions. Our project extends these analyses by considering for all weather hazards and displaying finer spatial resolution. Burke et al. (2021) found rising particulate pollution from wildfires, motivating the inclusion of diverse hazard types. Parks et al. (2023) estimated excess deaths after tropical cyclones, informing our approach to linking mortality and economic impact. In the context of hazard clustering and global comparisons, Seltzer and Kim-Sherman (2025) identified loss hot spots using national weather data, and Adil et al. (2025) ranked countries by a Climate Risk Index for weather-related damages. Arias et al. (2021) and CarbonBrief (n.d.) summarize global climate assessments, providing the broader scientific context. Finally, perception and livability studies such as U.S. News & World Report (2025) and World Economic Forum et al. (2022) underscore the value of integrating housing and relocation concerns. Our visualization incorporates all of these metrics and displays a similar ranking that's specific to the U.S. in order to inform users of the riskiest places to live currently, and the changing trends in weather-related risk.

4 Proposed Method

Innovations and Key Contributions:

Our method improves on current approaches through greater temporal granularity (monthly, county-level data spanning 25 years), unified normalization across heterogeneous domains (damages, aid, housing, and labor), and a visually interpretable framework that enables policymakers and the public to directly explore cause-effect patterns over time. We developed an experimental "county risk index" based on relative weather damage, FEMA spending, and economic shocks — offering a novel way to visualize counties' susceptibility to extreme weather. This index can easily incorporate new types of economic shocks, and is easy to update as new data becomes available. Our tool enables users to discover evolving patterns and dig into the details that comprise the risk index. By dynamically linking weather events, FEMA relief, and economic recovery metrics within an interactive D3.js map, our tool bridges the gap between quantitative climate indices and intuitive, actionable understanding — a step beyond prior static or single-hazard visualizations.

Data Collection:

- Census.gov provides county polygons for the year 2025. These form a GeoJSON file and allow us to plot all other attributes on a choropleth using D3.js.
- Census.gov also contains annual county populations from 2009 to present, and the decennial census in 2000 and 2010. These allow us to adjust dollar amounts per capita, which aids in county comparisons.
- The National Center for Environmental Information (NCEI) hosts data on severe weather events across the U.S. on its website. It contains the type and description of the storm event, the date and county affected, and the financial impact, injuries, and deaths attributed to the event. 1.3 million records totaling 3.4 Gb have been collected. We aggregate by county, month and event type. We then adjust dollar amounts for inflation, converting to 2025 value.
- FEMA data (979 MB) from 2000-present has been downloaded, and is specific to county and event date. We aggregate total public assistance obligations and individual assistance grants. We then adjust dollar amounts for inflation, converting to 2025 value
- Zillow data (102 MB) including housing market volatility, availability, and affordability metrics was downloaded, specific from 2000 - present. This data allows us to incorporate housing market recovery and depreciation into our analysis.
- Annual unemployment rates per county were downloaded from the Bureau of Labor Statistics (BLS).

Data Integration and Computation Pipeline: After extraction, all datasets are merged on county and time using FIPS codes and month-year. FIPS codes are mappable to state and county for data that does not already include them. Data wrangling is conducted in Python. Some weather events that are attributed to 'climate zones' instead of counties, so we use the center of the zone's geometry to map to the nearest county center. Event damages and FEMA assistance are inflation-adjusted using CPI, and normalized per capita using the nearest annual county population data from census.gov. The compiled final data is stored as CSV, which is split by year for faster loading in D3.

Defining Weather Risk Index: We construct a composite metric that reflects each county's relative damages, need for aid, and ability to withstand extreme weather each year. We filter the data each year to counties that experienced severe damages from weather (defined as greater than $10 per capita of damage in a single month). Other counties are automatically considered not risky that year. The index combines rankings across 4 dimensions: (1) **Exposure:** Total damage per capita from all storm events annually (from NCEI data). (2) **Aid Demand:** Total FEMA aid (related to storm events) per capita. (3) **Housing Market Shock:** Change in median home price 12 months after sufferiung damage from severe weather (from Zillow data). (4) **Labor Market Shock:** Increase in unemployment rate 12 months after suffering damage from severe weather (from BLS data). Annual county ranks are averaged to produce a final risk ranking index, and the top 100 per year are highlighted in our map.

So the final index is as follows: For each county that experienced damage from severe weather c in a given year t, we define:

$$R_{c,t} = \text{rank}\left(\frac{1}{4}\sum_{i=1}^{4} r_i(c,t)\right),$$

where $r_i(c,t)$ denotes the within-year rank of county c on (i) storm damage per capita, (ii) FEMA aid per capita, (iii) absolute percent housing price change, and (iv) percent unemployment increase.

Visualization and Interface Design: The visualization component uses a choropleth map built with D3.js, allowing dynamic filtering and time-based animation. Users can explore weather events by type, drill down into states and counties, and adjust the year slider to observe trends from 2000 to 2025. Hovering over counties will produce a popup summary of weather damages and FEMA spending, as well as a description of the most recent weather event.

5 Evaluation

To assess the effectiveness of our index, we examined the extent to which our average composite index score predicted disruption of Zillow's Market Heat Index (MHI), a proxy for housing market health, 12 months post severe weather event. We examined 600 county by year average risk scores that ranked among the 25 highest-risk counties in at least one year. A regression analysis showed a small, but statistically significant association between these risk scores and MHI change ($\beta = 0.09$, $t(598) = 9.75$, $p < 0.001$), suggesting that our risk ranking explains 13.7% of the change in MHI variance after a severe weather event ($R^2 = 0.137$). Future research should implement pre-storm control measures into this model as well as multiple time point assessments.

To provide further illustration of the real-word effectiveness of our risk ranking index and visualization, we examined counties with the highest risk rating during the catastrophic 2005 hurricane season, during which Hurricane Katrina caused widespread damage and disruption along the Louisiana coast and surrounding counties. Our visualization labels one of these gulf counties as the riskiest in the country for that year, and highlights many neighboring counties that also appear in the top 20 high-risk areas (Figure 1). This aligns with Burton (2015)'s case study findings, which showed limited recovery within the regions identified as the highest risk by our method in the year following Hurricane Katrina (Figure 2).

6 Conclusion And Discussion

We have successfully gathered and preprocessed relevant datasets from publicly available databases and built an interactive tool in D3.js. We derived a risk index using several components related to extreme weather events. We confirmed this index correlates as expected with its own inputs, that it's stable year over year, and responsive to shifting weather patterns in longer time frames. We discovered this index correlates with Zillow's Market Heat Index (MHI), which also aims to capture changes in local economies over time. We then performed a case study on hurricane Katrina: Fig. 1 this proved our map clearly displays major weather events and FEMA relief spending, and that our index properly captures the risk associated with a major disaster. Despite these promising findings, the overall association between our risk ranking and the MHI was low. This may suggest that the MHI is not an ideal proxy for resiliency and future risk, or that our index is lacking key features. Additionally, because Zillow Housing Data was included in the creation of the market heat index, some of the observed association may reflect shared method variance rather than meaningful variations. This highlights the need for additional validation of our measure across a broader variety of socioeconomic and recovery outcomes and time frames. In our initial project proposal, we intended to include insurance data. Due to time constraints and restricted access, we were unable to identify a robust dataset for inclusion. Future work should evaluate the benefit of incorporating such features on predictive accuracy.

Overall, the current study provides a strong foundation for future development. Moreover, it provides an intuitive and informative visualization that effectively communicates county-specific weather-related risk to both the public and government stakeholders. Such information can support a wide variety of applications, including helping individuals decide whereto live and identifying funding-aid disparities. Given the breadth of research predicting that severe weather events will continue to become more frequent and severe, this project highlights the critical need for ongoing investigation in this field.

Contributions:

1. Utsab Mitra: literature review, D3 development, index creation, and writeup
2. Sam Bolig: literature review, NCEI scraping and cleaning, index development, D3 development, and writeup
3. Katheryn Sauvigne: literature review, Zillow scraping and cleaning, index evaluation, and writeup
4. Daniel Damiano Cano: literature review and unemployment scraping
5. Jose Alejandro Rodriguez: literature review and unemployment scraping

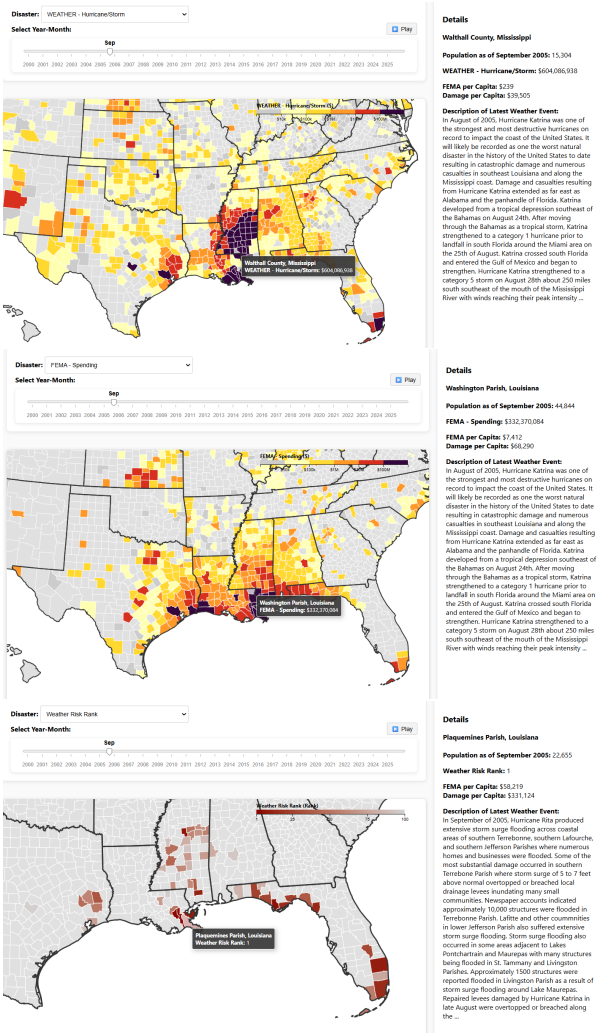

Figure 1: Case Study using Hurricane Katrina, 2005: (top) Storm Damage per capita, (middle) FEMA Funding per capita, (bottom) Risk Index.

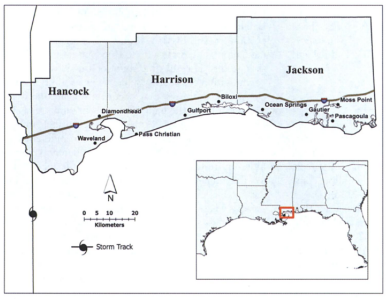

Figure 1. Study area. (Color figure available online.)

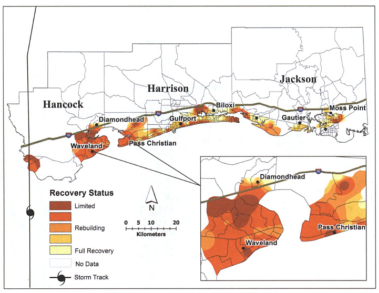

Figure 3. Recovery status for October 2005–October 2006. (Color figure available online.)

Figure 2: Limited recovery in same regions identified as highest risk by our visualization, adapted from Burton (2015).

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